

Project Title:

SlotMate

“Better Slots, Better Semester - Find your perfect class match”

Problem Statement:

In an open credit university system, students often struggle to get their desired class schedule due to student capacity, small time limit and network traffic. They may get unwanted sections, faculties or class times that are extremely difficult for them. Many students want to change their class section, but they do not find a suitable person to swap with. Most of the time, they depend on group chats or social media posts, which are messy, unorganized, and often get ignored.

There is no proper system to manage or match these requests and the whole process becomes slow, frustrating, and unreliable. Important messages get lost quickly and even when two students are willing to swap, they often do not find each other at the right time. As a result, many students miss out on possible swap opportunities and have to compromise with an unwanted schedule.

Project Idea:

SlotMate is a web-based platform that helps students find their desired class slots and best possible swap matches.

In that system, students will submit their request and instead of manual search the system will automatically match and connect them with others who have the opposite needs.

Project Overview:

START



User Login (University Email Verification)



Create Swap Request

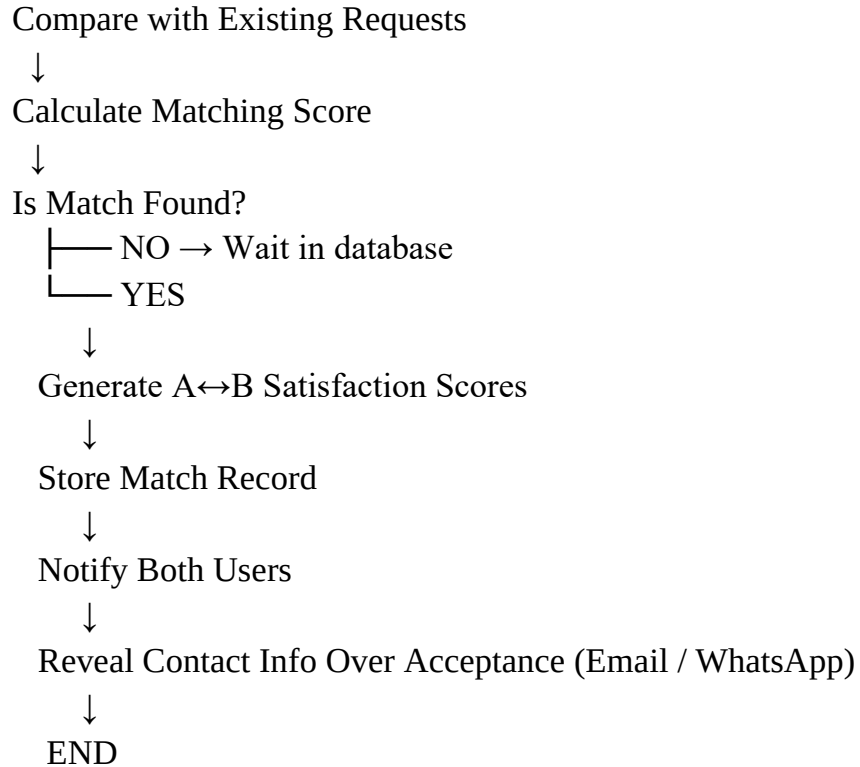


Store in Database



Trigger System Algorithm





Features:

1.Secure University Authentication

Users must register using:

- a) University Name
- b) University Email

2.Swap Request System

Users submit:

- a) Current section, faculty, time, days
- b) Preferred sections, faculties, time, days

3.Smart System Algorithm

- a) Weighted scoring (section, time, day, faculty)
- b) Bidirectional matching system

Shows:

A → B satisfaction %

B → A satisfaction %

4. Match Ranking System

- a) Perfect Match (80–100%)
- b) Strong Match (60–79%)
- c) Possible Match (40–59%)

5. Contact Sharing After Match

- a) No chat system
- b) Email or WhatsApp number is revealed only after the match acceptance. So, students communicate externally.

6. Real-Time Newsfeed Shows (Floating/Stable):

- a) New matching requests
- b) Available swaps & best match suggestions

7. Flexible Preference System (Optional)

- a) Any time / Any faculty option supported
- b) Range-based time selection (Morning/Afternoon/Evening)

Tools:

- 1. Frontend:** HTML, CSS, JavaScript
- 2. Backend:** Django
- 3. Database:** SQLite (Django default)
- 4. Matching Logic:** Python based weighted scoring algorithm in Django to compare swap requests and to find best matches
- 5. Version Control:** Git & GitHub (For project tracking and code management)
- 6. Development Tool:** VS code